西南财经大学本科期末考试试题册（A）

（2021－2022学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVA SE）(英)　命题教师：王磊

适用对象（年级专业）：2021 信息管理与信息系统专业

使用试题的任课教师姓名：王磊

试题说明：

1、考试类型：闭卷[ ]　开卷[√]

2、本套试题共 两 道大题，共 7 页，完卷时间 120 分钟。

3、考试用品中除纸、笔、尺子外，可另带的用具有：

计算器[ ]　字典[ ]　教材1本 [√]等

（请在下划线上填上具体数字或内容，所选[ ]内打钩）

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：　　　　**年级专业：**

学生姓名：　　　　**学　　号：**

考生注意事项：1.出示学生证（或身份证）和准考证于桌面左上角，以备查验。

2.拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。

3.答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。

4.所有答案均需填写在答题纸上，答在试题册上无效。

5.考生不得携带任何通讯工具进入考场。

6.考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。

7.严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

1. Which of these statements about constructors is **false**?

A. Its name must be the same as the class in which it is defined

B. There must be exactly one constructor defined for a class

C. Constructors are almost always declared as public

D. They can appear anywhere in the class where it is legal to declare a method

2. Which of the following Java statements set variable **even** to **true** if **n** is even, and to **false** if **n** is odd? (Assume n is an integer and satisfies n>=0.).

(hint: even numbers are those integers which, when divided by 2, have a remainder of 0.)

I. boolean **even** = (n/2.0 == (double)(n/2));

II. boolean **even** = (n % 2 == 0);

III. boolean **even** = (n/ 2 == 0);

IV. boolean **even** = (n % 2 == n/2);

A. I only

B. II only

C. I and II only

D. II and III only

3. What is the output of the following code fragment? Read it carefully!

```
String greet = "Hi";
String name = "Smedley";
String nickName = name.substring(0,4);
if (nickName == name.substring(0,4));
    System.out.println("has real nickname");
else if (greet + name == greet + nickName)
    System.out.println("no real nickname");
```

```
else
    System.out.println("hmmm...changed names?");
```

A. has real nickname

B. no real nickname

C. hmmm...changed names?

D. none, because there is at least one compile-time error

4. What is the output from the following code fragment?

```
String S = "", T = "";
int i = 4;
for (i = 1; i <= 3; i++)
    S = S + "!";

for (i = 1; i < 4; i=i+2)
    T = T + "*";
System.out.print(S);
System.out.println(T);
```

A. Nothing	B. !!!
C. !!!**	D. !!! ****

5. Which of the following statements about arrays are **false**?

I. Valid array indexes for an array called *myArray* are in the range between 1 and *myArray*.length inclusively.

II. Creating an array of object references (e.g. an array of String objects) does not create the actual objects whose references will be stored in the array.

III. Arrays themselves are objects, so an array is a reference to the actual array; therefore, two different array names can be used to refer to the same array in memory.

IV. If *myArray* is an array name, it is possible to make the array grow or shrink by changing the value of *myArray*.length.

A. I and III only B. I and IV only

C. II and IV only D. I and III

6. What is the output of the following program fragment?

```
public class A {
    public static int doStuff(double x, double y) {
        return (int)(x/y);
    }

    public static void main(String []args) {
        float x = 6.0;
        int y = 11;
        x = doStuff(y, x);
        System.out.print("x="+x+", y="+y);
    }
}
```

A. x=1, y=11

B. this program does not compile

C. x=6.0, y=11

D. x=1.0, y=11

7. If a class is not qualified as public or private, what does that imply about its public methods and fields?

A. No other class can use them: class access defaults to private.

B. Only classes in the same file can use them.

C. Only classes in the same package can use them.

D. Only classes derived from this class can use them.

8. An interface must meet which of these restrictions?

I. It must not have any fields.

II. All methods must be abstract.

III. Only public methods are allowed.

IV. A class can implement only one interface

A. II and III only B. I, III and IV only

C. I, II and III only D. I, II,III and IV

9. Given the following class definition of **Bird** and **Chicken**, which of the given statements will not compile? (hint: **Livestock** is an interface. We have omitted the contents in class body.)

abstract class **Bird** implements **Livestock** { … }

class **Chicken** extends **Bird** {…}

A. **Bird** bird = new **Chicken**( );

B. **Livestock** livestock = new **Chicken**();

(c) **Bird** bird = new **Bird**();

(d) None of these will compile

10. Given the following definitions of classes AA and BB. What is the output of the Test program?

```
public class Test2 {
        public static void main (String s[]) {
            AA refA = new AA();
            AA refB = new BB();
            refA.k();
            refB.k();
        }
}

class AA {
```

```
    protected void h( ) {
        System.out.print("h in classAA. ");
    }
    protected void g ( ) {
        System.out.print("g in classAA. ");
    }
    void k() {
        h();    g();
    }
}
class BB extends AA {
    protected void h() {
        System.out.print("h in classBB. ");
    }
    public void g() {
        System.out.print("g in classBB. ");
    }
    public void k() {
        h();     g();
    }
}
```

A. h in classAA. g in classAA. h in classBB. g in classBB.

B. h in classBB. g in classAA. h in classBB. g in classBB.

C. h in classBB. g in classBB. h in classBB. g in classBB.

D. h in classAA. g in classBB. h in classAA. g in classBB.

Part II: Write Programs.(total 70 pts)

1. (10pts) Complete the following **calcCylinder** method that will ask the user for the height and radius of a cylinder and then print out the volume of the cylinder ($2\pi r^2$ h). You may use the value 3.14159 or the constant Math.PI in your method.

```
public static double calcCylinder(double height,   double radius){


}
```

2. (10pts) Find the largest integer number **n** that satisfies $\sqrt{1} + 2 \times \sqrt{2} + 3 \times \sqrt{3} + 4 \times \sqrt{4} + \cdots + \mathrm{n} \times \sqrt{n} \leq 1000$.

3. (10pts) A number in the array is recognized as an outlier if it is greater than the mean 3 times. (e.g. 999 is recognized as an outlier in the below array)

```
int [ ] x= {4, 2, 10, 7, 1, 14, 3, 999, 5,12};
```

Complete the following method to find out the total number of outliers for the given array.

```
public static int countOutliers(int [ ] y) {
}
```

4. (20pts) Define an abstract class named **Object3D**, which contains:

- An member variable named **weight**, denoting the weight of the object
- An abstract method named **calcVolume** to calculate the volume of the object.
- An abstract method named **clacDensity** to calculate the density of the object.

Define a subclass **Cube** that extends the abstract class and contains:

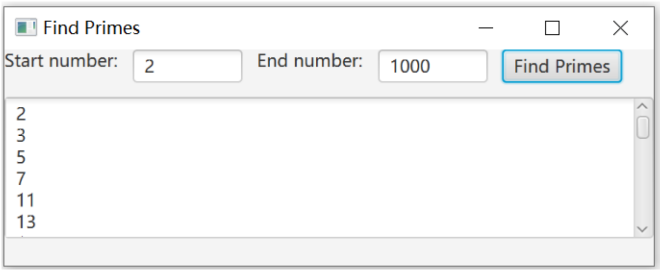
- A private member variable, **edge**, denoting the length of a cube object.
- A constructor.
- A **toString**( ) method to display information of the object, including: weight, edge, volume, density.

Define a subclass **Sphere** that extends the abstract class and contains:

- A private member variable, **radius**, denoting the radius of a sphere object.
- A constructor.
- A **toString**( ) method to display information of the object, including: weight, radius, volume, density.

In the **main** method, create a cube object and a sphere object, and output the information of the object having bigger density. (hint: the formula for calculating the volume of a sphere object is: $\frac{4}{3} \cdot \pi \cdot radius^3$

5. (20 pts) Write a GUI program to find out all the primes when the user gives a range and press down the button. The program may run like the below figure.



Requirements:

- Use suitable Java FX GUI components.
- Use suitable layout to organize components.
- At least contains one event handling process.